

Same Meaning, Different Answer?

Evaluating Answer Consistency in Large Language Models Across Linguistic Transformations

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Introduction & Research Gap

Background. LLMs are usually tested on questions with **fixed wording**. However, the same question can be written in different ways while keeping the same meaning and correct answer.

Research Gap. Previous work shows that rephrasing can affect model correctness (Srikanth et al., 2024). However, it is still important to understand how **different types of rephrasing** affect models across different reasoning tasks.

Our Study. We test three types of rephrasing: **lexical, syntactic, and information order**, across three reasoning tasks and three open-weight language models.

Research Questions & Hypothesis

RQ1 — Performance Change. How does model accuracy change after lexical, syntactic, and information-order transformations?

RQ2 — Correctness Stability. How often does a correct answer become incorrect, or an incorrect answer become correct, after rephrasing?

RQ3 — Robust Accuracy. How often do models answer correctly across all four versions of the same question?

Hypothesis. Rephrasing can change model correctness even when the meaning and correct answer stay the same. We expect this effect to differ across transformation types.

Contribution & Study Scope

The study compares meaning-preserving rephrasings across three reasoning tasks and three open-weight language models.

- **Datasets:** CommonsenseQA, GSM8K, LogiQA
- **Transformations:** Lexical, Syntactic, Information Order
- **Models:** Qwen2.5-3B, Llama-3.2-3B, Gemma-3-1B

30 questions × 4 versions × 3 models = 360 responses

Main Contribution. We compare each original question with three meaning-preserving versions of the same question. This paired evaluation measures **accuracy, correctness flips**, and **robust accuracy** to study how rephrasing affects model correctness.

Experimental Example

CommonsenseQA — Picture Frame

Original: Where can you put a picture frame when it's not hung vertically?

Lexical: Where can you place a picture frame when it is not hanging vertically?

Syntactic: When a picture frame is not hung vertically, where can it be placed?

Information Order: A picture frame is not being hung vertically. Where can you put it?

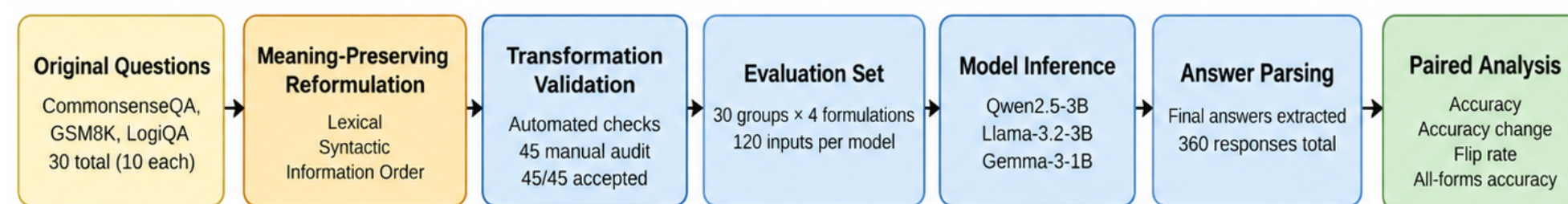
Correct Answer: E. Table

Key Point. The wording and structure change, but the meaning and correct answer remain the same.

Methodology & Experimental Design

Dataset Selection. We selected **30 questions**: 10 each from **CommonsenseQA**, **GSM8K**, and **LogiQA**, covering commonsense, mathematical, and logical reasoning.

Experimental Setup. Each question was tested in four versions: **Original**, **Lexical**, **Syntactic**, and **Information Order**. The transformed versions were designed to preserve the original meaning, reasoning requirements, and correct answer.



Transformation Types. **Lexical** changes replace words or phrases, **Syntactic** changes modify sentence structure, and **Information Order** changes rearrange existing facts while preserving the same meaning.

Paired Evaluation. Each original question and its three transformed versions form one question group. All four versions were tested on each model using the same settings, allowing us to compare correctness within the same question group.

Transformation Quality Control

All **90 transformations** passed structural checks to confirm that answer choices and numbers were preserved (where applicable) and that the text had changed.

We also manually reviewed a **stratified sample** to check that the original meaning and correct answer were preserved.

45 / 45 accepted

None of the reviewed transformations changed the correct answer, removed important information, or introduced new facts. This helps ensure that observed differences are more likely related to model behavior rather than invalid transformations.

Models & Evaluation

Models. We evaluated three open-weight instruction-tuned LLMs: **Qwen2.5-3B-Instruct**, **Llama-3.2-3B-Instruct**, and **Gemma-3-1B-IT**.

Inference. Each model was tested on **120 inputs** using deterministic decoding (do_sample=False, max_new_tokens=32). Only final answers were scored.

Evaluation Metrics.

- **Accuracy:** Percentage of correct answers.
- **Accuracy Change:** Transformed accuracy minus original accuracy.
- **Flip Rate:** Percentage of answers changing Correct↔Wrong.
- **Robust Accuracy:** Percentage correct across all four versions.

We measure consistency through **correctness**, not identical answer wording.

Statistics & Reproducibility

Statistical Analysis. We used exact **McNemar tests** to compare paired correctness, question-group **bootstrap 95% CIs** to estimate uncertainty, and **Benjamini–Hochberg correction** for multiple comparisons.

Reproducibility.

Code, data, validation, model outputs, and analysis are available at:

github.com/Yash-Gavade/ReasonShift



Main Result

Qwen: 63.3% → 30.0%

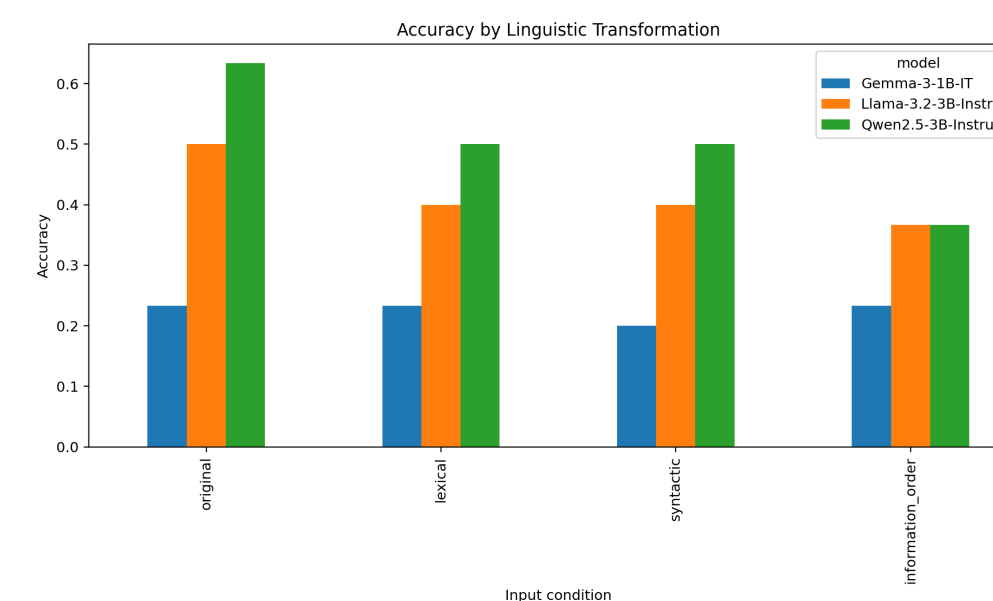
Original Accuracy → Robust Accuracy

Gap: –33.3 percentage points

Model	Original	Robust	Gap
Qwen	63.3%	30.0%	–33.3 pp
Llama	50.0%	26.7%	–23.3 pp
Gemma	23.3%	20.0%	–3.3 pp

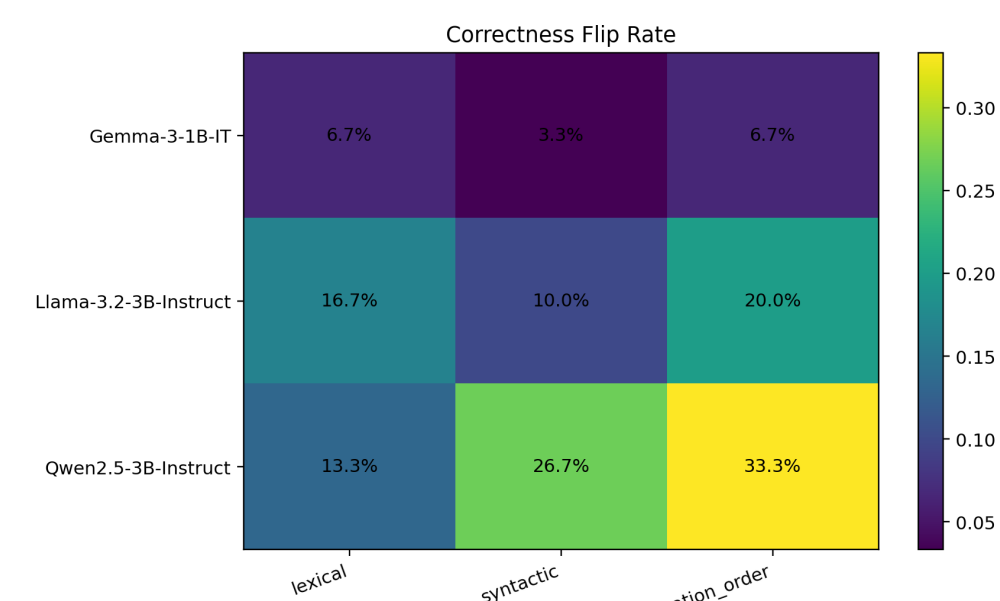
Key Result. All three models had lower robust accuracy than original accuracy. Qwen showed the largest gap at **33.3 percentage points**.

Results by Transformation



Accuracy by Transformation

Qwen: Original → Information Order
63.3% → 36.7%



Correctness Flip Rate

Qwen: Information Order
33.3% (9 C→W, 1 W→C)

Key Result. For Qwen, Information Order produced the largest accuracy drop, with **8 fewer correct answers out of 30**.

Findings & Hypothesis

- **Rephrasing changed correctness.** Some correct answers became incorrect, and some incorrect answers became correct.
- **Original accuracy missed instability.** All three models had lower robust accuracy, showing that performance changed across different versions of the same question.
- **Transformation type mattered.** For Qwen, Information Order showed the largest accuracy drop and highest flip rate.

Hypothesis. The results descriptively support our hypothesis, but none of the comparisons remained statistically significant after correction.

Discussion & Conclusion

Statistical Interpretation. For Qwen with Information Order, the exact McNemar test gave $p = 0.0215$, but after Benjamini–Hochberg correction this increased to $p_{BH} = 0.7734$. No comparison remained statistically significant after correction.

Limitations. The study used **30 question groups**, three small models due to computational limits, and three transformation types. Only **45/90 transformations** were manually checked for meaning preservation.

Conclusion. Rephrasing can change model correctness even when meaning is preserved. Testing multiple versions can reveal instability that original accuracy may miss. Larger studies with fully validated transformations are needed.